

Comprehensive AI Prompt for XplainCrypto Mobile App Development

Project Overview

Create a sophisticated mobile application that leverages XplainCrypto's Model Context Protocol (MCP) endpoints to deliver comprehensive cryptocurrency analysis, portfolio management, and research capabilities to users. The app should serve as a complete crypto analysis tool for both casual investors and professional analysts, with features for tracking portfolios, accessing real-time market data, generating detailed reports, and utilizing AI-powered insights.

Core Requirements

1. Develop a cross-platform mobile application (iOS and Android) that connects to XplainCrypto's MCP endpoints
2. Implement a clean, intuitive user interface following the mockups and design system provided
3. Create a secure user authentication system with multi-factor authentication options
4. Build comprehensive portfolio management capabilities with real-time valuation
5. Integrate real-time cryptocurrency market data from CoinGecko and CoinMarketCap endpoints
6. Incorporate DeFi analytics using DeFiLlama endpoints for protocol comparison and yield optimization
7. Implement research capabilities using Tavily endpoints for news aggregation and report generation
8. Develop AI-powered features using HuggingFace endpoints for personalized insights and a chatbot assistant
9. Design monetization features including subscription tiers and affiliate marketing opportunities
10. Ensure the app works offline with appropriate caching strategies and synchronization

XplainCrypto MCP Endpoints Integration

The application must integrate with the following XplainCrypto MCP endpoints:

CoinGecko Server

- `data://coingecko/{coin_id}` : Fetch comprehensive coin data
- `data://coingecko/price/{coin_id}` : Fetch price data
- `data://coingecko/market/{coin_id}` : Fetch market data
- Tool: `search_coins` : Search for coins by name or symbol

CoinMarketCap Server

- `data://coinmarketcap/{coin_id}` : Fetch comprehensive coin data
- `data://coinmarketcap/price/{coin_id}` : Fetch price data
- `data://coinmarketcap/market/{coin_id}` : Fetch market data
- Tool: `search_coins` : Search for coins by name or symbol

DeFiLlama Server

- `data://defillama/{protocol}` : Fetch TVL and protocol data
- `data://defillama/tvl/{protocol}` : Fetch Total Value Locked data
- `data://defillama/yields/{protocol}` : Fetch yields data
- Tool: `search_protocols` : Search for DeFi protocols by name

Tavily Server

- `data://tavily/search/{query}` : Fetch search results
- `data://tavily/news/{query}` : Fetch news results
- Tool: `search` : Search for information with adjustable search depth

HuggingFace Server

- `data://huggingface/embeddings/{text}` : Generate embeddings for text
- `data://huggingface/completion/{prompt}` : Generate text completions
- Tool: `infer_data` : Infer missing data based on query and data type

API Endpoints

- `POST /api/research` : Execute the research workflow for a given query
- `GET /api/tools` : Get all available tools from the specified MCP servers
- `POST /api/execute-tool` : Execute a specific MCP tool

- GET /api/initialize-endpoints : Manually reinitialize Pinecone with MCP endpoint metadata

Detailed Feature Requirements

1. User Authentication & Profile Management

Implement a secure authentication system with: - Email/password registration and login - Multi-factor authentication options (SMS, authenticator apps) - Biometric authentication (fingerprint, face recognition) - Social media login integration (optional) - User profile customization with avatars and preferences - Admin dashboard for user management and analytics

2. Market Data & Coin Information

Create comprehensive market data features including: - Cryptocurrency explorer with search and filtering - Real-time price updates using CoinGecko/CoinMarketCap endpoints - Detailed coin information pages with key metrics - Historical price charts with multiple timeframe options - Global market metrics dashboard - Trending coins based on price movement and social activity - Customizable price alerts with push notifications - Advanced charts with technical indicators and drawing tools

3. Portfolio Management

Develop robust portfolio tracking capabilities: - Multiple portfolio support with real-time valuation - Manual transaction entry and exchange API integration - Performance metrics and profit/loss calculations - Asset allocation visualization with pie charts and treemaps - Risk assessment based on volatility and correlation - Historical portfolio value charts - Transaction history with categorization and export options - Multiple watchlists with customizable metrics

4. DeFi Analytics & Tools

Implement DeFi-specific features using DeFiLlama endpoints: - Protocol explorer with TVL metrics and historical charts - Yield comparison across different protocols - Risk assessment for various DeFi platforms - Yield optimizer with APY/APR calculations and projections - DeFi portfolio tracking across multiple protocols - Protocol comparison with security and governance information

5. Research & Analysis

Create comprehensive research capabilities: - On-demand cryptocurrency research reports using XplainCrypto research workflow - Scheduled report generation for followed assets - Customizable report templates for different analysis types - News aggregator using Tavily endpoints with sentiment analysis - Social sentiment analysis for cryptocurrencies - On-chain analytics for transaction volume and network activity

6. AI-Powered Features

Develop intelligent features using HuggingFace endpoints: - Crypto Assistant chatbot with natural language query interface - Contextual awareness of market conditions and user portfolio - Personalized insights based on portfolio and market conditions - Predictive analytics for price trends and volatility - Content generation for market summaries and reports

7. Educational Resources

Include educational content and tools: - Learning hub with structured courses on cryptocurrency fundamentals - Interactive tutorials for using app features - Glossary of cryptocurrency terms - Strategy builder with risk assessment and backtesting - Simulation environment for paper trading

8. Monetization & Affiliate Features

Implement revenue-generating features: - Premium subscription tiers with different feature sets - Exchange integration with affiliate links - Marketplace for premium reports and strategies - Native advertising platform for crypto projects - Featured listings for new tokens

9. Utility Tools

Add practical tools for everyday crypto use: - Crypto calculator for currency conversion and investment planning - Calendar with economic events and token unlocks - Address book for frequently used wallet addresses - Notification center with customizable alerts - Data export options for accounting and taxes

UI/UX Requirements

The application should follow the UI mockups provided, with these key design principles:

- Clean and intuitive interface with minimalist design
- Data-focused layouts with emphasis on visualization
- Consistent design language across all screens
- Dark mode priority with light mode option
- Accessibility features for various user needs

Key screens to implement include: - Authentication screens (login, registration, onboarding) - Home and dashboard screens - Coin detail screens with advanced charts - Portfolio management screens - Research and analysis hub - DeFi analytics screens - AI assistant interface - Settings and profile screens

Technical Requirements

Architecture

- Follow the layered architecture outlined in the architecture document
- Implement clear separation between presentation, business logic, data, and network layers
- Use MVVM pattern for the presentation layer
- Implement repository pattern for data access

Performance

- Optimize for performance with efficient data structures
- Implement intelligent caching strategies for different data types
- Use background synchronization for non-critical updates
- Ensure smooth scrolling and responsive UI

Security

- Secure storage of user credentials and API keys
- Encryption of sensitive user data
- Secure communication with backend services
- Privacy controls for user data

Offline Support

- Core functionality available without internet connection
- Background synchronization when connectivity is restored
- Clear indication of data freshness and offline status

Monetization Strategy

The app should implement a tiered subscription model: - Basic (free): Limited market data, basic portfolio tracking - Pro (\$9.99/month): Advanced charts, unlimited portfolios, AI insights - Expert (\$19.99/month): All features, priority report generation, advanced AI tools - Institutional (\$99.99/month): API access, team accounts, custom reports

Additional revenue streams: - Affiliate marketing with cryptocurrency exchanges - Premium marketplace for reports and strategies - Native advertising for crypto projects and services

Development Guidelines

Technology Stack

- React Native or Flutter for cross-platform development
- Redux or MobX for state management
- Axios or similar for API communication
- Chart libraries optimized for financial data
- Secure storage libraries for sensitive information

API Integration

- Implement a robust API client for XplainCrypto endpoints
- Handle authentication and request signing
- Implement error handling and retry logic
- Optimize network usage with request batching and caching

Testing Requirements

- Comprehensive unit tests for business logic
- Integration tests for API communication
- UI tests for critical user flows
- Performance testing for data-intensive screens

Deployment

- CI/CD pipeline for automated builds and testing
- Beta testing program for early user feedback
- Phased rollout strategy for production

Additional Considerations

User Privacy

- Clear privacy policy explaining data usage
- Opt-in for analytics and personalization
- Data export and deletion options

Regulatory Compliance

- Disclaimer for investment advice
- Terms of service for data usage
- Compliance with financial regulations where applicable

Accessibility

- Support for screen readers
- Dynamic text sizing
- High contrast mode
- Voice command support where appropriate

Deliverables

The final deliverables should include: - Fully functional mobile application for iOS and Android - Source code with documentation - API integration documentation - User guide and help documentation - Admin dashboard for managing users and content

Timeline and Milestones

The development should follow these milestones: 1. Project setup and architecture implementation 2. Core functionality (authentication, market data, basic portfolio) 3. Advanced features (DeFi analytics, research, AI features) 4. UI polish and performance optimization 5. Testing and bug fixing 6. Beta release and user feedback 7. Final release and ongoing maintenance

Success Criteria

The project will be considered successful when: 1. The app successfully integrates with all XplainCrypto MCP endpoints 2. Users can manage portfolios, access market data, and generate reports 3. The AI features provide valuable insights and assistance 4. The app performs well on both iOS and Android devices 5. The monetization strategy generates revenue 6. User feedback is positive with high retention rates